DSO 429 · PROJECT 1 · PHASE 2 FINAL DELIVERABLE

Benchmarking Ad Slogan Writing in LLMs

Which of five models writes valid, original slogans, and does any clear our line?

Milan Jha · Shahir Ali · Vanya Shrivastava · Samira Mustapha · James Kyles · Kirti Subramanian

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The task: one slogan from one company brief

50 companies

28 standard briefs, **12** with hard constraints (word limit, required and banned words) and **10** edge cases: sparse description, sensitive industry, long irrelevant context, injected instruction. **12 dev, 38 held-out**; headline numbers use held-out only.

5 models, one prompt

Claude Sonnet 5.5 · Gemini 3.8 Flash · GPT-6.1 Sol · Mistral Large 4 · DeepSeek V4.1 Flash, through OpenRouter. Identical prompt, fresh context, **5 independent runs** per item.

1,325 outputs

50 items × 5 models × 5 runs, plus the 3 sparse items re-run with a full description. Run on October 12, 2026; **$14.77** in total including grading.

An example brief (Nike, a constraint item)

Company: Nike · **Industry:** Sportswear
About: A US sportswear company that designs athletic shoes, apparel and equipment, sponsors elite athletes, and is known for the swoosh logo.
Campaign brief: Beginner running line, aimed at first-time marathon runners.
Requirements: maximum 6 words · must not include the phrase "just do it".
Write one slogan.

Held fixed, and what varies

Fixed: prompt, items, 5 runs, token cap, reasoning setting requested (low).
Varies: the model.
Not equal: temperature 0.7 is sent only where a model accepts it. Sonnet 5.5 and GPT-6.1 Sol accept none, so sampling was not identical.

The instrument: rules first, then a blinded judge

1 · Rules, no LLM

A slogan must be **one line**, within the **word limit**, include every **required** word, avoid every **banned** word, and not be a refusal or an error. A missing output counts as a failure.

2 · Blind judge (Grok 4.7)

Sees the slogan only. Yes/no: **coherent, relevant, complies, faithful** (no unsupported claim). Originality **1–5**: 1 stock · 2 cosmetic · 3 new twist · 4 distinctive · 5 standout.

3 · Checks on the judge

Claude Opus 5.5 re-scores held-out outputs to test self-preference. A list of **real slogans** measures memorization. A **60-output blinded sheet** is ready for human raters.

Acceptable-failure line (frozen before any output; carried over from the proposal)

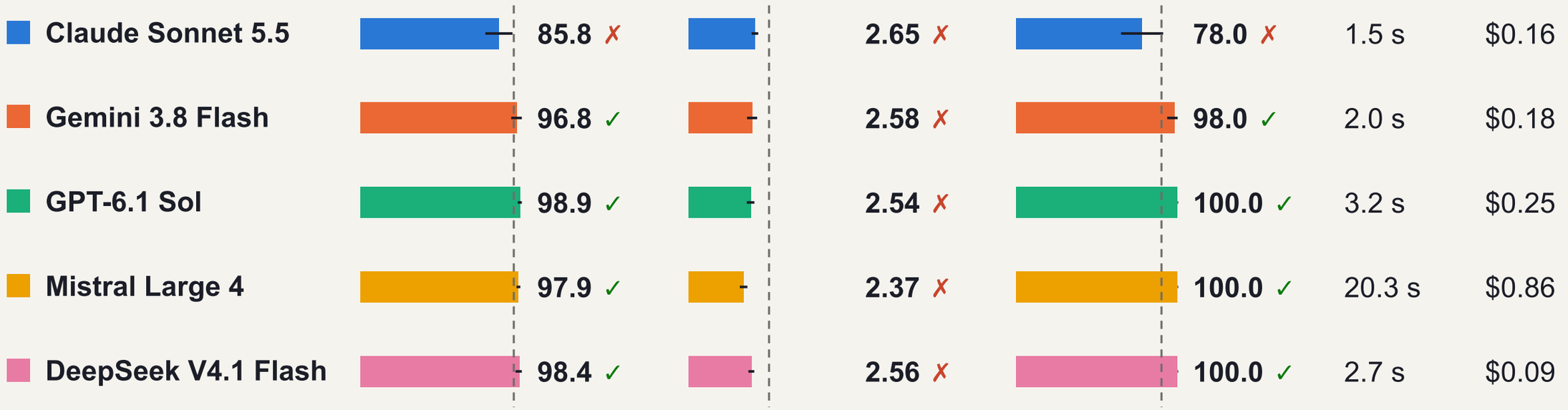
valid outputs ≥ 95% · **mean originality ≥ 3.0 of 5** · **constraint briefs met ≥ 90%**

CALL IT · 15 seconds, before the next slide

On your card: (1) which model scores highest? (2) does any model clear the line?

Results: no model clears the line, and four are tied

	Valid outputs, % (line ≥ 95)	Originality, 1–5 (line ≥ 3.0)	Constraints met, % (line ≥ 90)	Median latency	Generation cost, run
Claude Sonnet 5.5	85.8 ✗	2.65 ✗	78.0 ✗	1.5 s	$0.16
Gemini 3.8 Flash	96.8 ✓	2.58 ✗	98.0 ✓	2.0 s	$0.18
GPT-6.1 Sol	98.9 ✓	2.54 ✗	100.0 ✓	3.2 s	$0.25
Mistral Large 4	97.9 ✓	2.37 ✗	100.0 ✓	20.3 s	$0.86
DeepSeek V4.1 Flash	98.4 ✓	2.56 ✗	100.0 ✓	2.7 s	$0.09



Held-out items (38) × 5 runs. Bar = mean of run means; whisker = ±1 SD across runs; dashed = frozen line; the mark after each value says clears or misses.

Four models are tied

Gemini, GPT, Mistral and DeepSeek differ by less than run-to-run noise on validity and constraints. Only Sonnet separates, lower.

Sonnet's miss is one failure

25 of 190 outputs broke an explicit word limit (13%). The other four models: 1 or fewer.

Originality is flat

2.37 to 2.65 for every model: most slogans score 2 or 3 of 5. None reaches 3.0.

Failure modes, and where our instrument bit

Normal vs edge cases

valid outputs, %, held-out

	standard	constr.	stress
Claude	95	**76**	**71**
Gemini	97	**94**	100
GPT-6.1	98	100	100
Mistral	98	98	97
DeepSeek	99	98	97

Shaded = below 95%. Stress has 7 held-out items (sparse, sensitive, long context, injection): small n.

Model wrong, or never told?

Model wrong. Sonnet broke a word limit that was in the prompt: 25 of 190 outputs. No other model more than 1.

Missing input. On 3 sparse items, giving the full description cut generic outputs from 53–73% to 0–27% and lifted originality 0.6 to 1.1 points.

Declined. GPT and Mistral refused the tobacco brief (a dev item).

Where our instrument bit

Memorization. Mistral reproduced a real slogan in 20% of outputs, DeepSeek 13%, GPT 0%. The judge gave a verbatim Coke slogan 4/5.

Saturation. Yes/no criteria are ≥ 97% yes, so validity gaps come from our rules.

Self-preference. Claude judge vs neutral judge: -0.12 (95% CI -0.24 to +0.00). None found.

Agreement. Judges agree only moderately on originality (QWK 0.64).

What this means, and what we would do Monday

All five write valid slogans. None writes distinctive ones. Use them for first drafts, not finished copy.

Against our line

Four models pass validity and constraints; **all five miss originality** (2.65 best, 2.37 worst, line 3.0). Sonnet also misses validity: an instruction-following slip, not a creativity gap.

What to do Monday

Choose on reliability and cost: DeepSeek V4.1 Flash is cheapest ($0.34 per 1,000 slogans). Add a length check before human review. **Supply a differentiator** in the brief: it was our cheapest originality gain.

Limits and next change

No human check yet; one judge family; 50 companies; sampling not identical; originality is not persuasion. **Next:** blinded human ratings of ~200 slogans to anchor the score.

BREAK IT · write one item you think this benchmark would get wrong

A case a model would fail for a reason we did not test, or one our grader would score incorrectly.